

Extended Information:

**Scalable Patterning and Heterogeneous Integration of Diamond Quantum Microchips
in PIC and CMOS Foundry Processes**

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I. MODELING

To ensure reproducibility and transparency, we have developed a comprehensive software framework for the fabrication modeling and analysis presented in the main text. The code is organized into four main modules: (i) SEM image analysis for geometric parameter extraction, (ii) FDTD electromagnetic simulations, (iii) sensitivity analysis and surrogate modeling, and (iv) inverse Monte Carlo for thickness estimation. The complete codebase is available at github.com/abuzzimit/Photonic-Crystal-Cavity-Simulations.

A. SEM Analysis

The extraction of geometric parameters from scanning electron microscope (SEM) images is performed using a multi-stage computer vision pipeline implemented in Python with OpenCV and scikit-image. We analyze both top-view images (for waveguide width and hole radius) and tilted images (for waveguide thickness estimation).

1. Image Preprocessing

Raw SEM images undergo bilateral filtering (diameter $d = 9$, $\sigma_{\text{color}} = 75$, $\sigma_{\text{space}} = 75$) to reduce noise while preserving edge sharpness, followed by non-local means denoising. Edge detection is performed using the Canny algorithm with adaptive thresholds ($T_{\text{low}} = 10$, $T_{\text{high}} = 60$) optimized for the contrast characteristics of diamond nanostructures imaged at 3 kV accelerating voltage.

2. Waveguide Width Extraction from Top-View Images

The waveguide rails are detected using a gradient-based approach combined with principal component analysis (PCA) for robust orientation estimation. The structure tensor method provides the global tangent angle θ of the nanobeam:

$$\theta = \frac{1}{2} \arctan \left(\frac{2\langle J_{xy} \rangle}{\langle J_{xx} \rangle - \langle J_{yy} \rangle} \right) \quad (1)$$

where $J_{ij} = G_i G_j$ are the structure tensor components computed from Scharr gradient operators G_x, G_y .

Along the direction $\mathbf{m} = (-\sin \theta, \cos \theta)$ perpendicular to the waveguide axis, we compute the signed gradient $g_{\perp} = G_x m_x + G_y m_y$ and construct histograms to identify the waveguide edges (Supplementary Fig. 1). The positive histogram $H_+(s)$ accumulates values where $g_{\perp} > 0$ (bright-to-dark transitions), while the negative histogram $H_-(s)$ accumulates values where $g_{\perp} < 0$. The trapezoidal waveguide cross-section produces four distinct peaks corresponding to the bottom left, top left, top right, and bottom right edges. For width extraction, we use the outermost opposite-polarity peaks (bottom left and bottom right), which correspond to the bottom width of the trapezoidal cross-section.

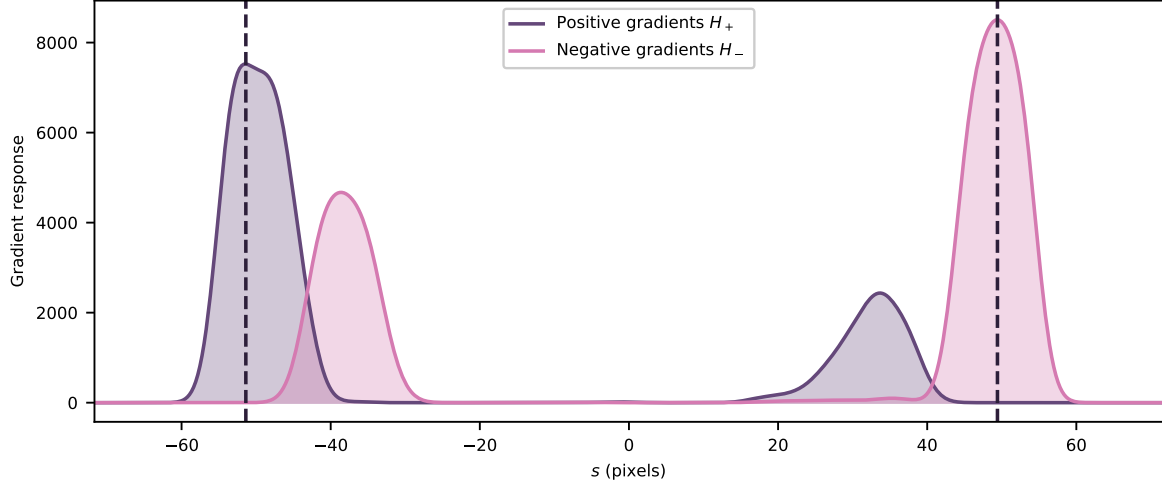
Candidate rail positions along the normal direction are refined using iteratively-reweighted least squares (IRLS) with Cauchy-type robust weighting:

$$w_i = \frac{w_i^{(0)}}{1 + (r_i/6\sigma_{\text{MAD}})^2} \quad (2)$$

where the initial weights $w_i^{(0)} = |g_{\perp}|_i$ are the gradient magnitudes at each sample point, r_i is the residual, and σ_{MAD} is the median absolute deviation. The rail separation $W = |d_2 - d_1|$ is converted to nanometers using the calibrated scale factor (3.39 nm/px).

3. Hole Radius Extraction

Circular holes within the photonic crystal region are detected using a combination of Hough circle detection and contour analysis. The detection is constrained to the central strip between the rails, defined by $|\mathbf{x} \cdot \mathbf{m} - d_{\text{mid}}| <$



Supplementary Fig. 1. Signed gradient histograms for waveguide rail detection. The positive gradient histogram H_+ (purple) and negative gradient histogram H_- (pink) are plotted as functions of position s along the normal direction to the waveguide axis. The four peaks, from left to right, correspond to the bottom left, top left, top right, and bottom right edges of the trapezoidal waveguide cross-section.

$0.45 \cdot W$ where $d_{\text{mid}} = (d_1 + d_2)/2$ is the midline position. Detected circles are filtered to retain only those with radii in the range $30 \leq r \leq 70$ nm, centers within 8% of the rail separation from the midline, radii consistent within $8\sigma_{\text{MAD}}$ of the median, and spacing greater than 110 nm between adjacent holes along the tangent direction.

B. Thickness Extraction from Tilted SEM Imagery

To provide an independent validation of the thickness values inferred from resonance wavelength modeling, we directly measure the cavity thickness from 45° tilted SEM images by analyzing the projected trapezoidal cross-section. The measurement pipeline consists of four main stages: (i) geometric modeling of the projection, (ii) image preprocessing and edge detection, (iii) robust line fitting and labeling, and (iv) scale-free thickness calculation with uncertainty propagation.

1. Geometric Model and Projection

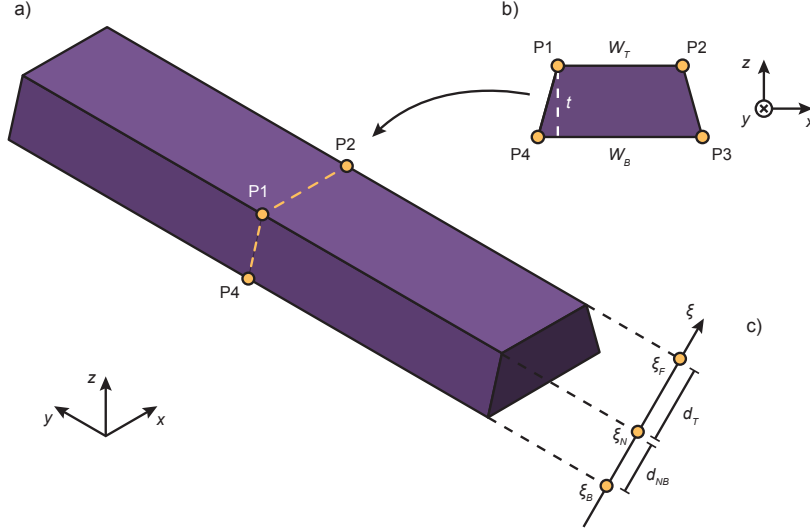
Coordinate systems. Consider a nanobeam waveguide with trapezoidal cross-section characterized by three parameters: top width W_t , bottom width W_b , and thickness t (Supplementary Fig. 2a–b). The nanobeam is oriented along the y -axis (length direction), with its cross-section lying in the xz -plane. The top surface is at $z = +t/2$ and the bottom surface at $z = -t/2$.

We define three coordinate systems: (i) *sample coordinates* (x, y, z) fixed to the nanobeam with z pointing upward, y along the nanobeam axis, and x perpendicular to both (Supplementary Fig. 2a); (ii) *camera coordinates* (u, v, w) fixed to the SEM detector with w along the electron beam direction and (u, v) forming the image plane; and (iii) *image coordinates* (X, Y) as the 2D projection onto the detector in pixels. The detection algorithm measures edge positions along the *normal coordinate* ξ , which is perpendicular to the detected ridge direction.

Rotation and projection. The SEM stage tilts the sample by angle θ about the y -axis. Additionally, the nanobeam may have an in-plane rotation (azimuth) ϕ relative to the tilt axis. The tilt rotation R_θ about the y -axis and the azimuthal rotation R_ϕ about the z -axis are:

$$R_\theta = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}, \quad R_\phi = \begin{pmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (3)$$

A point $\mathbf{r} = (x, y, z)^T$ in sample coordinates transforms to camera coordinates via $\mathbf{r}' = R_\theta \cdot R_\phi \cdot \mathbf{r}$. The SEM operates



Supplementary Fig. 2. Geometric model of the trapezoidal nanobeam cross-section and its projection under tilt. a) 3D view of the trapezoidal nanobeam. b) Cross-section showing the trapezoidal geometry with dimensions W_t (top width), W_b (bottom width), and t (thickness) noted. c) Projection along the ξ direction (normal to the ridge direction) showing the three visible edges and the separations d_T and d_{NB} used for thickness extraction.

in orthographic projection mode (parallel rays), so the projected position along the camera u -axis is:

$$u = sx', \quad x' = x \cos \theta \cos \phi - y \cos \theta \sin \phi + z \sin \theta, \quad (4)$$

where s is the pixel scale (px/nm). The term $-y \cos \theta \sin \phi$ is common to all points at the same y_0 and cancels in all inter-edge distances.

Visible edges at 45° tilt. At a fixed longitudinal coordinate $y = y_0$, the trapezoidal cross-section has four vertices:

$$\begin{aligned} P_1 &= \left(-\frac{W_t}{2}, y_0, +\frac{t}{2} \right), & P_2 &= \left(+\frac{W_t}{2}, y_0, +\frac{t}{2} \right), \\ P_3 &= \left(+\frac{W_b}{2}, y_0, -\frac{t}{2} \right), & P_4 &= \left(-\frac{W_b}{2}, y_0, -\frac{t}{2} \right). \end{aligned} \quad (5)$$

At 45° tilt, three horizontal edges are visible: both top edges (P_1 and P_2) and one bottom edge. The far bottom edge (P_3) is occluded by the body of the nanobeam; thus the visible bottom edge corresponds to the near-sidewall bottom ridge (P_4). The projected positions along the camera u -axis are:

$$u_F = s \left(+\frac{W_t}{2} \cos \theta \cos \phi + \frac{t}{2} \sin \theta \right) \quad (\text{far top, from } P_2), \quad (6)$$

$$u_N = s \left(-\frac{W_t}{2} \cos \theta \cos \phi + \frac{t}{2} \sin \theta \right) \quad (\text{near top, from } P_1), \quad (7)$$

$$u_B = s \left(-\frac{W_b}{2} \cos \theta \cos \phi - \frac{t}{2} \sin \theta \right) \quad (\text{near bottom, from } P_4). \quad (8)$$

Separations along normal coordinate. The projected positions u_F , u_N , and u_B defined above are measured along the camera u -axis (the x' component after rotation). However, the detection algorithm measures edge positions along the normal coordinate ξ (perpendicular to the ridge direction), as illustrated in Supplementary Fig. 2c. After rotation by $-\psi$ to align the nanobeam horizontally, the normal coordinate ξ is rotated relative to the camera u -axis.

Since we only need the *separations* between edges (differences in coordinates), and both d_T and d_{NB} are measured along the same normal coordinate system, their ratio d_{NB}/d_T is invariant under this coordinate transformation. This is why the scale-free thickness formula works: it depends only on the ratio of separations, not on the absolute coordinate system.

We denote the normal coordinate positions as ξ_F , ξ_N , and ξ_B to distinguish them from the camera u -axis positions u_F , u_N , and u_B . The separations in ξ are proportional to the separations in u (related by a rotation factor), but since we only use ratios, the exact relationship is not needed. The key relationships are:

The *top-top separation* along the normal coordinate is:

$$d_T \equiv \xi_F - \xi_N = s W_t \cos \theta \cos \phi, \quad (9)$$

where ξ_F and ξ_N are the normal coordinate positions of the far top and near top edges, respectively.

The *near-top to near-bottom separation* along the normal coordinate is:

$$d_{NB} \equiv \xi_N - \xi_B = s \left(\frac{W_b - W_t}{2} \cos \theta \cos \phi + t \sin \theta \right), \quad (10)$$

where ξ_B is the normal coordinate position of the near bottom edge. Note the $(W_b - W_t)/2$ term arises because both ξ_N and ξ_B are on the same (near) sidewall.

2. Thickness Inversion Formula

Derivation. Rearranging Eq. (10) to isolate the thickness:

$$d_{NB} = s \frac{W_b - W_t}{2} \cos \theta \cos \phi + s t \sin \theta \quad (11)$$

$$s t \sin \theta = d_{NB} - s \frac{W_b - W_t}{2} \cos \theta \cos \phi \quad (12)$$

yielding

$$t = \frac{1}{\sin \theta} \left(\frac{d_{NB}}{s} - \frac{W_b - W_t}{2} \cos \theta \cos \phi \right). \quad (13)$$

Scale-free form. To eliminate the pixel scale s , we use Eq. (9):

$$s = \frac{d_T}{W_t \cos \theta \cos \phi}. \quad (14)$$

Substituting into Eq. (13):

$$t = \frac{1}{\sin \theta} \left(\frac{d_{NB} \cdot W_t \cos \theta \cos \phi}{d_T} - \frac{W_b - W_t}{2} \cos \theta \cos \phi \right) \quad (15)$$

$$= \frac{\cos \theta \cos \phi}{\sin \theta} \left(\frac{W_t d_{NB}}{d_T} - \frac{W_b - W_t}{2} \right) \quad (16)$$

giving the scale-free formula:

$$t = \cot \theta \cos \phi \left(\frac{W_t d_{NB}}{d_T} - \frac{W_b - W_t}{2} \right). \quad (17)$$

Special cases. For $\phi = 0$ (nanobeam aligned with the tilt axis):

$$t = \cot \theta \left(\frac{W_t d_{NB}}{d_T} - \frac{W_b - W_t}{2} \right). \quad (18)$$

For $\phi = 0$ and $\theta = 45^\circ$ ($\cot \theta = 1$):

$$t = \frac{W_t d_{NB}}{d_T} - \frac{W_b - W_t}{2}. \quad (19)$$

3. Azimuth Angle Correction

When the nanobeam axis is not aligned with the tilt axis ($\phi \neq 0$), the azimuth angle appears foreshortened in the tilted image. We derive the relationship between the apparent image angle ψ and the true azimuth ϕ .

Derivation. Consider a unit vector along the nanobeam axis in the sample plane:

$$\mathbf{v} = (\sin \phi, \cos \phi, 0). \quad (20)$$

Under tilt θ about the y-axis, the rotation matrix is

$$R_\theta = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}. \quad (21)$$

The rotated vector is:

$$\mathbf{v}' = R_\theta \mathbf{v} = (\sin \phi \cos \theta, \cos \phi, -\sin \phi \sin \theta). \quad (22)$$

In orthographic projection, the image coordinates are $(v'_x, v'_y) = (\sin \phi \cos \theta, \cos \phi)$. The apparent angle in the image is:

$$\tan \psi = \frac{v'_x}{v'_y} = \frac{\sin \phi \cos \theta}{\cos \phi} = \tan \phi \cos \theta. \quad (23)$$

Solving for the true azimuth:

$$\boxed{\phi = \arctan \left(\frac{\tan \psi}{\cos \theta} \right)}. \quad (24)$$

Expressing $\cos \phi$ in terms of ψ . Starting from

$$\tan \phi = \frac{\tan \psi}{\cos \theta}, \quad (25)$$

we write $\tan^2 \phi$ and use $1 + \tan^2 \phi = \sec^2 \phi = 1/\cos^2 \phi$:

$$\tan^2 \phi = \frac{\tan^2 \psi}{\cos^2 \theta} \implies \cos^2 \phi = \frac{1}{1 + \tan^2 \phi} = \frac{1}{1 + \frac{\tan^2 \psi}{\cos^2 \theta}} = \frac{\cos^2 \theta}{\cos^2 \theta + \tan^2 \psi}. \quad (26)$$

Using $\tan^2 \psi = \sin^2 \psi / \cos^2 \psi$ and multiplying numerator and denominator by $\cos^2 \psi$ gives

$$\cos^2 \phi = \frac{\cos^2 \theta \cos^2 \psi}{\cos^2 \theta \cos^2 \psi + \sin^2 \psi}. \quad (27)$$

Finally, rewrite the denominator using $\sin^2 \psi = 1 - \cos^2 \psi$:

$$\begin{aligned} \cos^2 \theta \cos^2 \psi + \sin^2 \psi &= \cos^2 \theta \cos^2 \psi + 1 - \cos^2 \psi \\ &= 1 - \sin^2 \theta \cos^2 \psi, \end{aligned} \quad (28)$$

so that

$$\boxed{\cos \phi = \frac{\cos \theta \cos \psi}{\sqrt{1 - \sin^2 \theta \cos^2 \psi}}} \quad (29)$$

where we take the positive square root since $|\phi| < 90^\circ$ in our geometry (so $\cos \phi > 0$).

Thickness formula in terms of measured quantities. Substituting Eq. (29) into Eq. (17):

$$\boxed{t = \frac{\cos^2 \theta \cos \psi}{\sin \theta \sqrt{1 - \sin^2 \theta \cos^2 \psi}} \left(\frac{W_t d_{NB}}{d_T} - \frac{W_b - W_t}{2} \right)}. \quad (30)$$

Simplification for $\theta = 45^\circ$. At $\theta = 45^\circ$, we have $\sin \theta = \cos \theta = 1/\sqrt{2}$ and $\sin^2 \theta = 1/2$. The prefactor becomes:

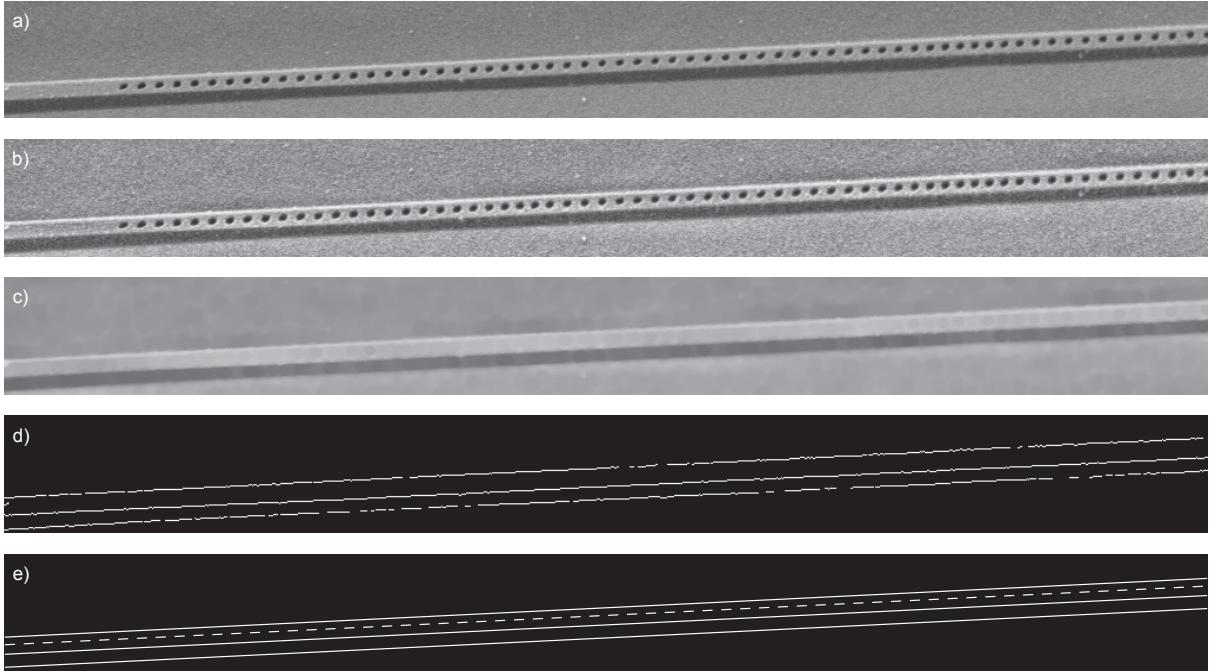
$$\frac{\cos^2 \theta \cos \psi}{\sin \theta \sqrt{1 - \sin^2 \theta \cos^2 \psi}} = \frac{(1/2) \cos \psi}{(1/\sqrt{2}) \sqrt{1 - \frac{1}{2} \cos^2 \psi}} = \frac{\cos \psi}{\sqrt{2 - \cos^2 \psi}} = \frac{\cos \psi}{\sqrt{1 + \sin^2 \psi}}. \quad (31)$$

Thus for $\theta = 45^\circ$:

$$t = \frac{\cos \psi}{\sqrt{1 + \sin^2 \psi}} \left(\frac{W_t d_{NB}}{d_T} - \frac{W_b - W_t}{2} \right). \quad (32)$$

For small apparent angles ($|\psi| \lesssim 5^\circ$), this reduces to $t \approx W_t d_{NB}/d_T - (W_b - W_t)/2$.

4. Image Preprocessing and Edge Detection



Supplementary Fig. 3. Tilted SEM image preprocessing and edge detection pipeline. a) Original 45° tilted SEM image of the nanobeam cavity. b) After CLAHE contrast enhancement, showing improved local contrast. c) After hole filling, with morphological closing. d) Canny edge detection produces a binary edge map highlighting ridge boundaries. e) Detected line segments from the probabilistic Hough transform (solid lines) and the reconstructed hidden edge (dashed line), showing the three identified ridge edges used for thickness measurement.

Preprocessing pipeline. Raw SEM images undergo a multi-stage preprocessing pipeline optimized for diamond nanostructures imaged at 3 kV accelerating voltage (Supplementary Fig. 3a). The pipeline consists of four sequential operations:

1. **Contrast enhancement:** Adaptive histogram equalization (CLAHE) with clip limit 2.0 and tile grid size 8×8 pixels improves local contrast while preventing over-amplification of noise (Supplementary Fig. 3b).
2. **Edge-preserving denoising:** Bilateral filtering (diameter $d = 9$, $\sigma_{\text{color}} = 75$, $\sigma_{\text{space}} = 75$) reduces noise while preserving edge sharpness by weighting pixels based on both spatial proximity and intensity similarity. This step is applied after CLAHE but is not shown as a separate panel in the figure.
3. **Hole filling:** Morphological closing with an elliptical structuring element of size 9×9 pixels fills small dark holes and pits in the surface before edge detection (Supplementary Fig. 3c). This prevents surface defects from creating spurious edges that would interfere with ridge detection.

4. **Edge detection:** Canny edge detection with thresholds $T_{\text{low}} = 140$ and $T_{\text{high}} = 150$, aperture size 3, and L2 gradient norm, produces a binary edge map (Supplementary Fig. 3d).

Line segment detection. Line segments are detected using the probabilistic Hough transform (HoughLinesP) with the following parameters: $\rho = 1$ pixel, $\theta = \pi/180$ rad resolution, threshold 40 votes, minimum line length 200 pixels, and maximum line gap 20 pixels. These parameters are tuned to detect the long, nearly horizontal ridge edges while rejecting short spurious segments from surface roughness. The detected line segments (solid lines) and the reconstructed hidden edge (dashed line) are shown overlaid on the original image in Supplementary Fig. 3e, displaying the three identified ridge edges (far top, near top, and near bottom) used for thickness measurement.

5. Robust Line Fitting and Orientation Estimation

Grouping by intercept. Detected line segments are grouped by their y-intercept b in the line equation $y = mx + b$. For each segment with endpoints (x_1, y_1) and (x_2, y_2) , the slope and intercept are computed as $m = (y_2 - y_1)/(x_2 - x_1)$ and $b = y_1 - mx_1$. Segments with $|\Delta x| < 5$ pixels (near-vertical) are discarded. Segments are sorted by intercept and clustered using a tolerance of 4 pixels, ensuring that segments belonging to the same physical ridge are grouped together.

Robust line fitting. For each group of segments, a single line is fitted to all segment endpoints using OpenCV's `fitLine` function with the Welsch robust loss function (DIST_WELSCH). This loss function is less sensitive to outliers than least squares, preventing stray segments from pulling the fitted line away from the true ridge position. The fit returns a line representation (v_x, v_y, x_0, y_0) where (v_x, v_y) is the unit direction vector and (x_0, y_0) is a point on the line.

Orientation estimation. The global orientation angle ψ of the nanobeam in the image is determined from the fitted lines using circular statistics. For each fitted line with angle α_i (computed as $\arctan 2(v_y, v_x)$), we compute the mean angle:

$$\psi = \arctan 2 \left(\frac{1}{N} \sum_{i=1}^N \sin \alpha_i, \frac{1}{N} \sum_{i=1}^N \cos \alpha_i \right), \quad (33)$$

where N is the number of detected lines. This approach correctly handles the circular nature of angles (e.g., 359° and 1° average to 0° , not 180°).

Line selection and sorting. The top $n = 3$ line groups are selected based on total segment length (sum of lengths of all segments in each group), prioritizing the longest, most prominent ridges. The selected lines are sorted by y-intercept (top to bottom) to establish a consistent ordering: $\xi_F > \xi_N > \xi_B$.

6. Distance Calculation and Line Labeling

Perpendicular distance between two parallel lines. Consider two parallel lines with common slope m ,

$$\ell_1 : y = mx + b_1, \quad \ell_2 : y = mx + b_2. \quad (34)$$

Writing them in implicit form $mx - y + b_i = 0$, both lines share the same normal vector $\mathbf{n} = (m, -1)$ with $\|\mathbf{n}\| = \sqrt{1 + m^2}$. Choose points $P_1 = (0, b_1) \in \ell_1$ and $P_2 = (0, b_2) \in \ell_2$, so that $\Delta \mathbf{r} = P_2 - P_1 = (0, b_2 - b_1)$. The separation along the normal direction equals the magnitude of the projection of $\Delta \mathbf{r}$ onto the unit normal $\hat{\mathbf{n}} = \mathbf{n}/\|\mathbf{n}\|$:

$$d = |\Delta \mathbf{r} \cdot \hat{\mathbf{n}}| = \left| (0, b_2 - b_1) \cdot \frac{(m, -1)}{\sqrt{1 + m^2}} \right| = \frac{|b_2 - b_1|}{\sqrt{1 + m^2}}. \quad (35)$$

Line labeling with plausibility scoring. Given three detected lines sorted by y-intercept, we must assign labels: far top (ξ_F), near top (ξ_N), and near bottom (ξ_B). Since the lines are already sorted top-to-bottom, the physical ordering constraint is $\xi_F > \xi_N > \xi_B$ (with the topmost line being ξ_F and the bottommost being ξ_B).

For the single valid assignment (indices 0, 1, 2), we compute the separations using the perpendicular distance formula derived above, where the y-intercepts b_i correspond to the normal coordinate positions ξ_i of the fitted lines:

$$d_T = \frac{|\xi_N - \xi_F|}{\sqrt{1 + m^2}}, \quad d_{NB} = \frac{|\xi_B - \xi_N|}{\sqrt{1 + m^2}}, \quad (36)$$

where $m = \tan \psi$ is the common slope of the lines.

The thickness is computed using Eq. (30) or Eq. (32), and a plausibility score is assigned based on:

- **Range check:** Thickness must lie within physically plausible bounds ($t_{\min} \leq |t| \leq t_{\max}$, typically 20–400 nm). Violations are penalized.
- **Sign preference:** Positive thickness is preferred (negative thickness indicates incorrect labeling).
- **Ratio check:** The ratio $r = d_{NB}/d_T$ should be reasonable (typically $0.1 < r < 5.0$). Extreme ratios suggest measurement errors.
- **Regularization:** A small penalty proportional to $|t|$ prevents selection of unrealistically large thickness values.

The assignment with the lowest score is selected. In practice, with properly sorted lines, only one assignment needs to be evaluated, but the scoring framework provides robustness against edge cases.

7. Uncertainty Propagation

Monte Carlo uncertainty estimation. Uncertainties in the measured widths W_t and W_b (obtained from top-view SEM analysis) are propagated to thickness estimates using Monte Carlo sampling. The procedure consists of four steps:

Step 1: Input distributions. Width measurements are characterized by their sample statistics:

$$W_t \sim \mathcal{N}(\bar{W}_t, \sigma_{W_t}) \quad (37)$$

$$W_b \sim \mathcal{N}(\bar{W}_b, \sigma_{W_b}) \quad (38)$$

where $\bar{W}_t, \bar{W}_b$ are the mean values and $\sigma_{W_t}, \sigma_{W_b}$ are the standard deviations.

Step 2: Forward sampling. For each Monte Carlo iteration $j = 1, \dots, N_{MC}$ (typically $N_{MC} = 2000$):

1. Draw $W_t^{(j)} \sim \mathcal{N}(\bar{W}_t, \sigma_{W_t})$
2. Draw $W_b^{(j)} \sim \mathcal{N}(\bar{W}_b, \sigma_{W_b})$
3. Clip values to ensure $W_t^{(j)} > 0$ and $W_b^{(j)} > 0$

Step 3: Thickness computation. For each sample $(W_t^{(j)}, W_b^{(j)})$, compute the thickness using the scale-free formula:

$$t^{(j)} = \cot \theta \cos \phi \left(\frac{W_t^{(j)} d_{NB}}{d_T} - \frac{W_b^{(j)} - W_t^{(j)}}{2} \right), \quad (39)$$

where d_T and d_{NB} are the measured separations (held fixed), θ is the tilt angle, and ϕ is the recovered azimuth angle.

Step 4: Output statistics. The thickness distribution is characterized by:

$$\bar{t} = \frac{1}{N_{MC}} \sum_{j=1}^{N_{MC}} t^{(j)} \quad (40)$$

$$\sigma_t = \sqrt{\frac{1}{N_{MC} - 1} \sum_{j=1}^{N_{MC}} (t^{(j)} - \bar{t})^2} \quad (41)$$

Percentiles (5th, 50th, 95th) are also computed to provide confidence intervals. This approach naturally captures correlations and nonlinearities in the projection geometry without requiring analytical error propagation.

For a representative cavity measurement with top width $W_t = 238$ nm and bottom width $W_b = 314$ nm, the Monte Carlo uncertainty analysis ($N_{MC} = 2000$ samples) yields a thickness distribution with mean $\bar{t} = 143.4$ nm and standard deviation $\sigma_t = 13.6$ nm.

Validation. The thickness distribution obtained from this direct measurement method is validated against the thickness values inferred from resonance wavelength modeling (see Inverse Monte Carlo section). Agreement between the two independent methods confirms the validity of both the geometric model and the parameter extraction pipeline.

C. FDTD Simulations

Finite-difference time-domain (FDTD) simulations were performed using Tidy3D, a cloud-accelerated electromagnetic solver. The simulation framework handles geometry import from GDS files, material dispersion, and comprehensive monitor setup.

1. Material Model

Diamond optical properties are modeled using the two-term Sellmeier dispersion relation:

$$n^2(\lambda) = 1 + \frac{B_1\lambda^2}{\lambda^2 - C_1} + \frac{B_2\lambda^2}{\lambda^2 - C_2} \quad (42)$$

with coefficients $B_1 = 0.3306$, $C_1 = (0.175 \mu\text{m})^2$, $B_2 = 4.3356$, $C_2 = (0.106 \mu\text{m})^2$, valid for wavelengths 0.23–5 μm . At the SnV emission wavelength of $\lambda = 620$ nm, this yields $n = 2.424$.

2. Simulation Configuration

The photonic crystal cavity geometry is imported directly from foundry GDS layout files, ensuring that the simulated structure exactly matches the fabricated device layout. The simulation domain is sized to encompass the cavity region and sufficient surrounding space to capture the evanescent field decay. Perfectly matched layer (PML) boundary conditions are applied on all six faces to absorb outgoing waves and prevent spurious reflections from domain boundaries. The PML uses the default Tidy3D configuration with 8 layers, which provides effective absorption for the wavelength range of interest.

The excitation source is a point dipole with E_y polarization positioned at the cavity center, driven by a Gaussian pulse with center frequency corresponding to $\lambda_0 = 620$ nm and relative bandwidth $\Delta f/f_0 = 12\%$. Grid resolution is set to a minimum of 18 steps per wavelength, automatically refined in high-index regions.

3. Q-Factor Analysis

Cavity quality factors are extracted from time-domain field ringdown data using the Tidy3D ResonanceFinder plugin based on Prony's method. A point field monitor records $E_y(t)$ at the cavity center with temporal resolution of 5 simulation time steps.

The analysis identifies resonant modes by decomposing the field into a sum of decaying sinusoids:

$$E(t) = \sum_j A_j e^{(i\omega_j - \alpha_j)t} \quad (43)$$

where ω_j is the angular frequency, α_j is the decay rate, and the quality factor is:

$$Q_j = \frac{\pi f_j}{\alpha_j} = \frac{\omega_j}{2\alpha_j} \quad (44)$$

Mode selection prioritizes resonances with $1000 < Q < 500,000$ and selects the highest-amplitude mode within this range to identify the fundamental cavity resonance.

D. Sensitivity Analysis and Surrogate Model

To construct a surrogate model relating geometric parameters to resonance wavelength, we perform a systematic sensitivity analysis using FDTD simulations.

1. Parameter Space Sampling

The three-dimensional parameter space (waveguide width W , hole radius r , cavity thickness t) is sampled using a 32-point Sobol quasi-random sequence within $\pm 30\%$ of nominal values. Sobol sequences provide superior space-filling properties compared to random sampling, ensuring uniform coverage of the parameter space with fewer simulation points.

For each parameter combination (W_i, r_i, t_i) , an FDTD simulation determines the resonance wavelength $\lambda_{\text{res},i}$.

2. Surrogate Model Construction

The surrogate model is a quadratic polynomial fit capturing linear effects, interactions, and curvature:

$$\lambda_{\text{res}}(\mathbf{x}) = \beta_0 + \sum_{k=1}^3 \beta_k x_k + \sum_{k \leq l} \beta_{kl} x_k x_l \quad (45)$$

where $\mathbf{x} = (x_W, x_r, x_t)^T$ are the normalized geometric parameters:

$$x_k = \frac{p_k - p_k^{(0)}}{\Delta p_k} \quad (46)$$

with $p_k^{(0)}$ the nominal value and Δp_k the variation range.

The coefficients $\beta = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_{11}, \beta_{12}, \beta_{13}, \beta_{22}, \beta_{23}, \beta_{33})^T$ (10 coefficients total: 1 constant, 3 linear, 6 quadratic terms) are determined by ordinary least squares:

$$\beta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\lambda} \quad (47)$$

where $\mathbf{X}$ is the $N \times 10$ design matrix with N rows (one per sample point) and 10 columns (one per polynomial term). For sample point i with normalized parameters $(x_W^{(i)}, x_r^{(i)}, x_t^{(i)})$, row i of $\mathbf{X}$ contains the 10 polynomial terms evaluated at that point: $[1, x_W^{(i)}, x_r^{(i)}, x_t^{(i)}, (x_W^{(i)})^2, x_W^{(i)} x_r^{(i)}, x_W^{(i)} x_t^{(i)}, (x_r^{(i)})^2, x_r^{(i)} x_t^{(i)}, (x_t^{(i)})^2]$. The vector $\boldsymbol{\lambda}$ contains the corresponding resonance wavelengths $\lambda_{\text{res},i}$ from the FDTD simulations.

3. Model Validation

Cross-validation against held-out FDTD simulations confirms a mean absolute error of < 1 nm in resonance wavelength prediction. The sensitivity coefficients $\partial \lambda_{\text{res}} / \partial p_k$ extracted from the surrogate model quantify the relative importance of each geometric parameter.

E. Inverse Monte Carlo

The inverse problem of estimating cavity thickness from measured resonance wavelengths is solved using Monte Carlo sampling combined with the surrogate model.

1. Problem Formulation

Given:

- Measured resonance wavelength distribution $\{\lambda_{\text{meas},i}\}$ from hyperspectral cavity characterization
- Measured waveguide width distribution $\{W_i\}$ from top-view SEM analysis
- Measured hole radius distribution $\{r_i\}$ from top-view SEM analysis
- Surrogate model $\lambda_{\text{res}}(W, r, t)$

Find: The thickness distribution $\{t_i\}$ consistent with the observations.

2. Monte Carlo Sampling Procedure

Step 1: Input distributions. The experimentally measured distributions are characterized by their sample statistics:

$$W \sim \mathcal{N}(\bar{W}, \sigma_W) \quad (48)$$

$$r \sim \mathcal{N}(\bar{r}, \sigma_r) \quad (49)$$

$$\lambda_{\text{meas}} \sim \text{Empirical distribution from measurements} \quad (50)$$

Step 2: Forward sampling. For each Monte Carlo iteration $j = 1, \dots, N_{\text{MC}}$:

1. Draw $W_j \sim \mathcal{N}(\bar{W}, \sigma_W)$
2. Draw $r_j \sim \mathcal{N}(\bar{r}, \sigma_r)$
3. Draw λ_j from the measured wavelength distribution

Step 3: Surrogate model inversion. For each sample (W_j, r_j, λ_j) , solve for thickness t_j by inverting the surrogate model:

$$t_j = \lambda_{\text{res}}^{-1}(\lambda_j; W_j, r_j) \quad (51)$$

For the quadratic surrogate model, this inversion is performed analytically. Rearranging the surrogate equation into the standard quadratic form $at^2 + bt + c = 0$, where $a = \beta_{33}$, $b = \beta_3 + \beta_{13}x_W + \beta_{23}x_r$, and $c = \beta_0 + \beta_1x_W + \beta_2x_r + \beta_{11}x_W^2 + \beta_{12}x_Wx_r + \beta_{22}x_r^2 - \lambda_j$, we solve for t_j using the quadratic formula. The physically meaningful root (positive and within the expected thickness range) is selected.

Step 4: Output statistics. The thickness distribution is characterized by:

$$\bar{t} = \frac{1}{N_{\text{MC}}} \sum_{j=1}^{N_{\text{MC}}} t_j \quad (52)$$

$$\sigma_t = \sqrt{\frac{1}{N_{\text{MC}} - 1} \sum_{j=1}^{N_{\text{MC}}} (t_j - \bar{t})^2} \quad (53)$$

3. Results

The inverse Monte Carlo analysis was performed with $N_{\text{MC}} = 10,000$ samples using the following input parameters: width $W = 0.330 \pm 0.004 \mu\text{m}$, radius $r = 0.045 \pm 0.004 \mu\text{m}$, and resonance wavelength $\lambda = 633.2 \pm 9.0 \text{ nm}$. The resulting thickness distribution has mean $\bar{t} = 0.129 \mu\text{m}$ and standard deviation $\sigma_t = 0.012 \mu\text{m}$.

These results are consistent with the thickness values obtained from direct measurement using tilted SEM imagery (see Section II.B), validating both the geometric model and the parameter extraction pipeline.